

Course: B. Tech.

Semester : VI

Branch : Electronics & Telecommunication Engineering and Allied

Subject Code & Name: Antennas and Wave Propagation [BTETC601]

Max Marks: 60

Date: 18/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)	CO01,02,03	12
1 Which of the following equations describes the power density of an electromagnetic wave in free space?		1
a. Friis Transmission Equation b. Maxwell's Equations		
c. Poynting Vector Equation d. Wave Equation		
2 Which of the following frequencies is typically used for ground wave propagation?		1
a. VHF b. UHF c. Microwave d. LF		
3 What is the MUF?		1
a. The highest frequency that can be used for sky wave propagation		
b. The lowest frequency that can be used for sky wave propagation		
c. The frequency at which the ionosphere is most reflective		
d. The frequency at which the ionosphere is least reflective		
4 Which of the following is NOT a type of antenna?		1
a. Dipole antenna b. Parabolic reflector		
c. Capacitor d. Horn antenna		
5 What is the main difference between a broadside array and an end-fire array?		1
a. The direction of maximum radiation		
b. The number of elements in the array		
c. The spacing between the elements		
d. The polarization of the elements		
6 Which of the following is a parasitic element in a Yagi-Uda antenna array?		1
a. The driven element b. The reflector		
c. The director d. Both reflector and director		

Q. 4 Solve Any Two of the following. **12**

- A. Explain the two element uniform array and prove that **CO 02** **6**

$$E = 2E_A \cos\left(\frac{\pi d \cos \phi}{\lambda} + \frac{\alpha_e}{2}\right)$$

- B. Discuss the principle of pattern multiplication in an N-element linear array. How does the radiation pattern evolve as the number of elements increases? **CO 03** **6**

- C. Compare the characteristics of a binomial and Dolph Tchebyshev antenna array. Explain the design principles behind each type and how they affect the antenna's performance. **CO 03** **6**

Q.5 Solve Any Two of the following. **12**

- A. Compare adaptive and fixed beamforming techniques along with the distinguishing features of each, and in what scenarios are they most effective? **CO 02** **6**

- B. Explain MUF. A high frequency radio link has to be established between two points as a distance of 2700 km on earth's surface. Consider the ionospheric height to be 250 km and its critical frequency 5 MHz, calculate the MUF for the given path **CO 02** **6**

- C. Explain the advantages of smart antennas in wireless communication systems **CO 03** **6**

Q. 6 Solve Any Two of the following. **12**

- A. Explain travelling wave Antennas in detail. **CO 03** **6**

- B. Explain the structural design and working principle of antennas with parabolic reflectors. Discuss the radiation pattern and the focusing properties of such antennas. **CO 03** **6**

- C. Write a short note on (a) Rhombic Antenna and (b) V antenna. **CO 02,03** **6**

***** End *****